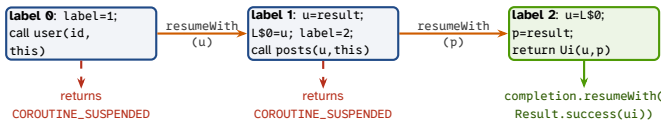


In one line: The compiler rewrites every suspend fun in **continuation-passing style**: it gains a hidden Continuation parameter, returns Any?, and its body becomes a **state machine** whose Label says where to resume. A suspension returns COROUTINE_SUSPENDED all the way up and frees the thread; later something calls resumeWith, the **dispatcher** picks a thread, the machine jumps to the next label. Job, scopes, cancellation = *library* (kotlinx.coroutines) on top.

```
suspend fun load(id){ val u = user(id); val p = posts(u); return Ui(u,p) }
```

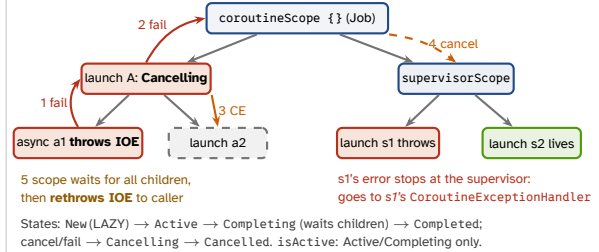
Load\$1 : ContinuationImpl
label, result, L\$0 (spilled u), completion

one object per call, reused for every resume
invokeSuspend(result) = switch(Label)



resume: callback thread → resumeWith → DispatchedContinuation → dispatcher.dispatch() → invokeSuspend on the chosen thread. Fast path: callee returned a value, not the marker → fall through.

Job tree — failure up, cancellation down



The CPS transform

- suspend fun f(x: A): T → JVM Object f(A x, Continuation<? super T> c). Return is T or the marker COROUTINE_SUSPENDED (kotlin.coroutines.intrinsics) — hence Any?.
- interface Continuation<in T> { val context: CoroutineContext; fun resumeWith(result: Result<T>) } — a typed callback carrying its context.
- Only functions that call suspend functions get a state machine; one ContinuationImpl subclass per function, **locals live across a suspension are spilled** into fields (L\$0, I\$0... (unverified)), the rest stay on the stack. A suspended coroutine = that heap object, **no thread held**.

Context & dispatch

- CoroutineContext = **indexed set** of Elements keyed by Key: Job, ContinuationInterceptor (the dispatcher), CoroutineName, CoroutineExceptionHandler. a + b: the right side wins per key; ctx[Job], minusKey.
- A child's context = parent context + builder arg + a **new child Job**. Passing Job() as an argument **breaks the parent link**.
- Dispatcher = interceptor: interceptContinuation wraps so resume → dispatch().
- Default: #cores (min 2), CPU work. IO: 64 or #cores, whichever is larger — **shares threads with Default**, so withContext(IO) from Default often does not switch thread. IO.limitedParallelism(n) is *elastic* (beyond the 64); Default.limitedParallelism(n) is a view within Default. Main.immediate: no re-post if already on main.

Example — bridge a callback, cancel cleanly

```

suspend fun Call.await(): Response =
    suspendCancellableCoroutine { cont ->
        cont.invokeOnCancellation { cancel() } // propagate inward
        enqueue(object : Callback {
            override fun onResponse(c: Call, r: Response) = cont.resume(r)
            override fun onFailure(c: Call, e: IOException) =
                cont.resumeWithException(e)
        })
    }
suspend fun crunch(xs: List<Int>) = withContext(Dispatchers.Default) {
    for (x in xs) { ensureActive(); heavy(x) } // cooperative
}
try { work() } finally {
    withContext(NonCancellable) { db.flush() } // may suspend
}
runCatching { fetch() } // TRAP: also catches CancellationException
.onFailure { if (it is CancellationException) throw it }
  
```

Structured concurrency & cancellation

- Rules: every coroutine has a parent Job; a parent **completes only after its children**; **cancelling a parent cancels all children**; a child failing with a non-CancellationException **cancels its parent** and so its siblings.
- job.cancel() takes only a CancellationException = *normal*: parent **not** cancelled, handlers ignore it.
- Cancellation is **cooperative**: a flag that bites at the next *cancellable* suspension point (delay, yield, withContext, await...) by throwing CancellationException. CPU loops check isActive/ensureActive()/yield().
- Suspending in finally after cancel throws again → withContext(NonCancellable). withTimeout throws a CE subclass.

Exceptions — who sees what

Where	What happens to an exception
root launch	uncaught → its CoroutineExceptionHandler, else the thread's uncaught handler (Android: crash)
root async any child	stored in the Deferred; thrown at await(); a CEH on it has no effect delegated to the parent, up to the root — a CEH on a child is never used ; an un-awaited child async still fails the parent
coroutineScope	cancels the rest, waits, rethrows to the caller — a normal try/catch around it works
supervisorScope / SupervisorJob	child failure is not propagated; direct children use <i>their own</i> CEH (like roots)
several fail	first exception wins, the rest are suppressed

Traps

- !** runCatching/catch (e: Exception) swallows CancellationException: the coroutine runs on after cancel.
- !** launch(SupervisorJob()) in a scope: that Job becomes the parent — nothing supervised, scope cancel no longer reaches it.
- !** try/catch around launch {} catches nothing — the exception goes up the Job tree, not up the call stack.
- !** Blocking calls ignore cancel — use runInterruptible.

Remember: suspend = callback + state machine · failure up, cancel down · CE is normal · handler at the root.